

2a	The increase in internal energy ΔU of a system is equal to the sum of the heat supplied Q to the system and the work done W on the system.
2bi	$\text{Work done on gas} = -p(V_2 - V_1)$ $= -(1.04 \times 10^5)(4.0 \times 10^{-4} - 2.9 \times 10^{-4})$ $= -11.4 \text{ J}$ (1 mark if answer is not negative)
2bii	$\Delta U = Q + W$ $= 28.5 + (-11.4)$ $= 17.1 \text{ J}$
2biii	$\frac{V}{T} = \frac{V'}{T'}$ $T' = \frac{V'}{V} T = \frac{4.00 \times 10^{-4}}{2.9 \times 10^{-4}} (314)$ $= 433 \text{ K}$
2biv	$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$ $\sqrt{\langle c^2 \rangle} = \sqrt{3kT/m}$ Since k and m are constant, $\text{Ratio} = \sqrt{T_f/T_i} = \sqrt{433/314} = 1.17$

2c	A transverse wave is a wave in which the displacement of the particles in the wave are
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